

FALL 2026



EE3003 Analog and Digital Communications

Instructor: Mohsin Yousuf
Department of Electrical Engineering

FALL 2026

Course Learning Outcomes

CLO No.	Course Learning Outcome (CLO) Statements	Assessment Tools	Taxonomy Levels	PLO
2	Calculate parameters like bandwidth, power efficiency, transmission efficiency, and modulation index for linear (AM, SSB, VSB, QAM) and non-linear (FM, PM) analog modulation systems.	Q2, M1, M2	C3	1

FALL 2026

Lecture Objectives

- Why Modulation?
- Modulation Categories
- Baseband Vs Carrier Communication
- Amplitude Modulation
 - DSB-SC
 - Coherent Demodulation

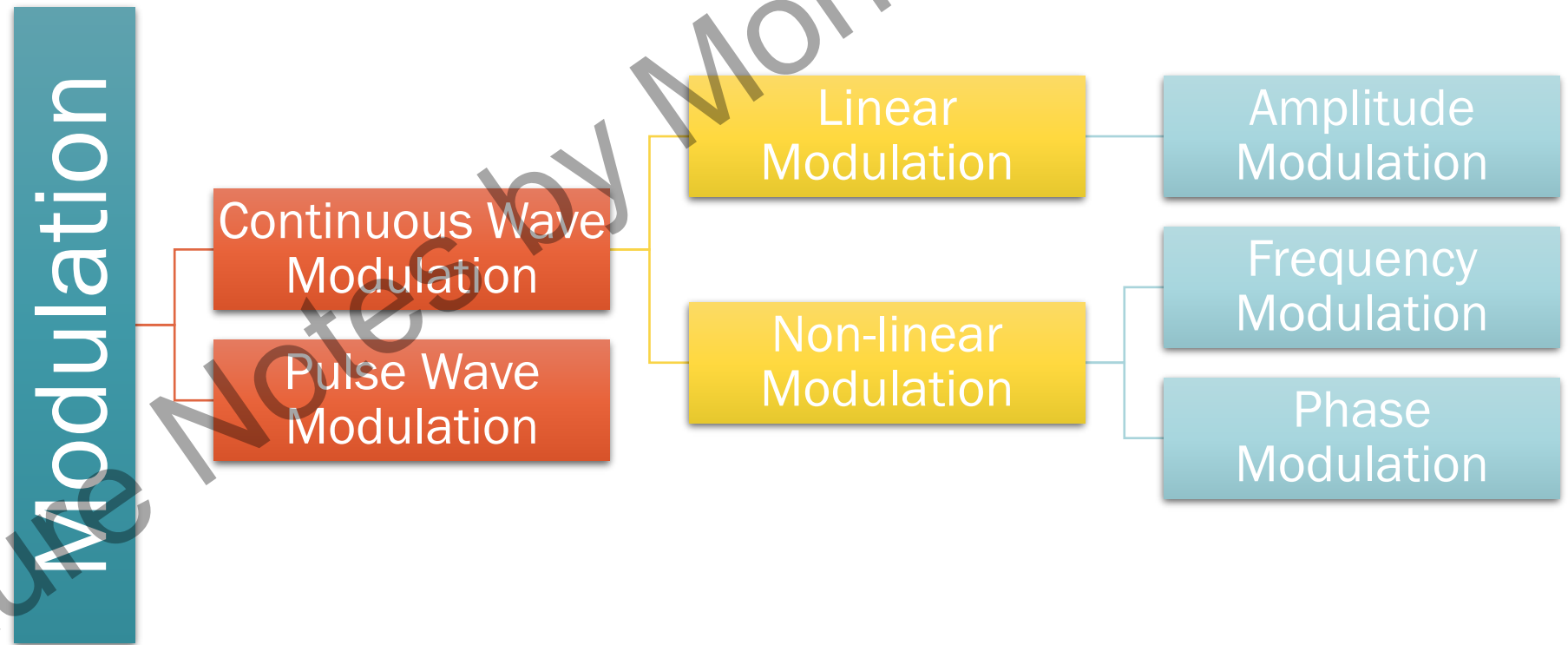
Why Use Modulation?

- Most input signals, often created by **transducers**, can't be sent directly over the communication channel. These signals are baseband signals (i.e., information, message or data) and would overlap each other if simultaneously communicated.
- Rather, a **carrier wave**, whose properties are better suited to the **transmission requirements**, is conditioned (i.e., **modulated**) to contain the information of the signal.
- **Modulation** is the systematic alteration of a **carrier wave** in order to “carry” the information, message or data to be communicated.

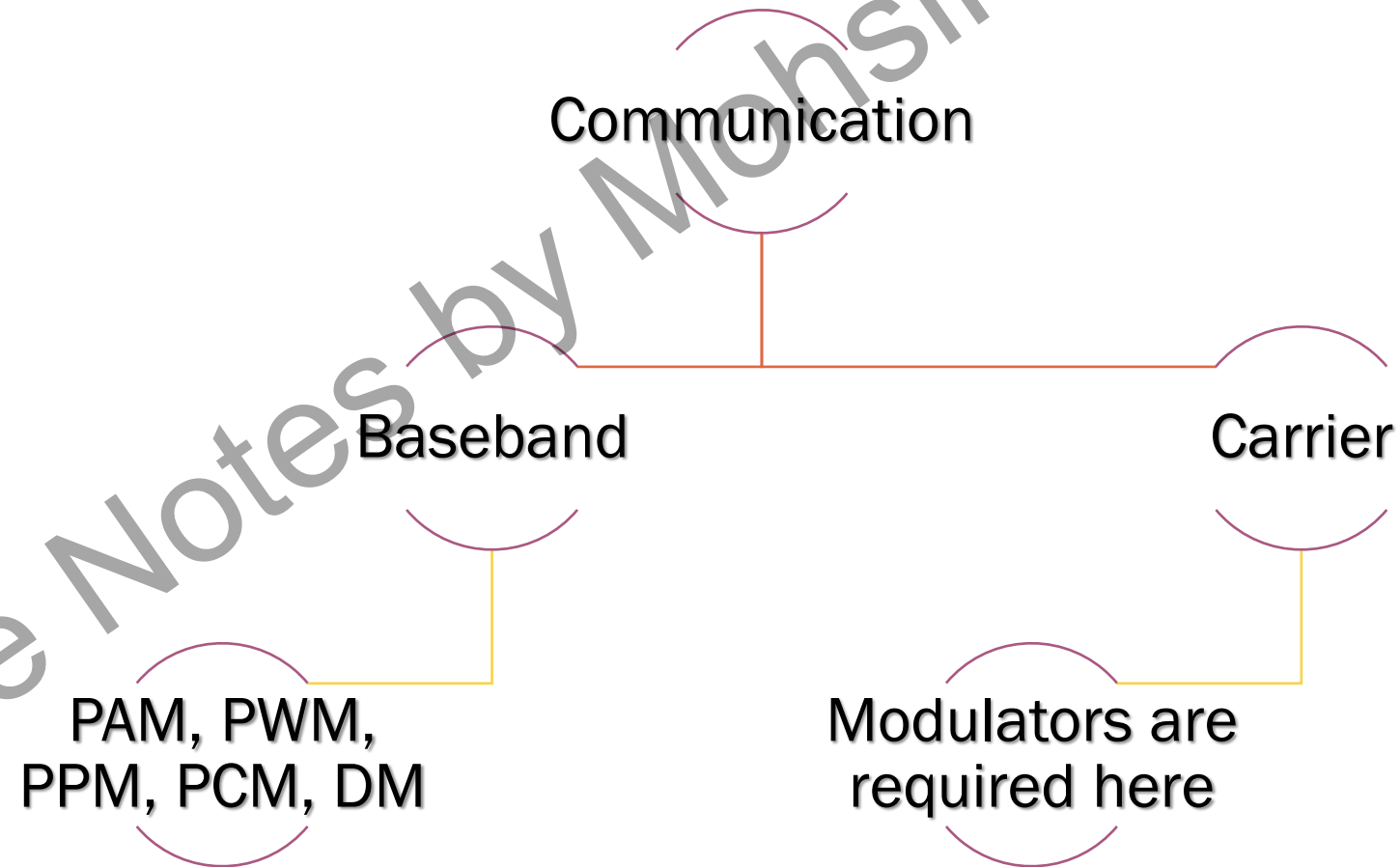
Why Use Modulation?

- Modulation for Smaller Antennas
 - ❖ Antennas must be greater than one-tenth of the wavelength ($\lambda/10$); thus low frequency baseband signals would require overly large antennas.
- Modulation for frequency assignment
 - ❖ The PTA assigns frequency bands to each radio application. So, different carrier frequencies are used to meet this need.
- Modulation for multiplexing
 - ❖ This allows for multiple signals to be carried on a single transmission medium (multiplexing is one form of modulation).
- Modulation to exchange SNR and bandwidth B
 - ❖ As was the case in power-limited or band-limited channels. Some types of modulation are useful for reducing noise and interference.

Modulation Categories

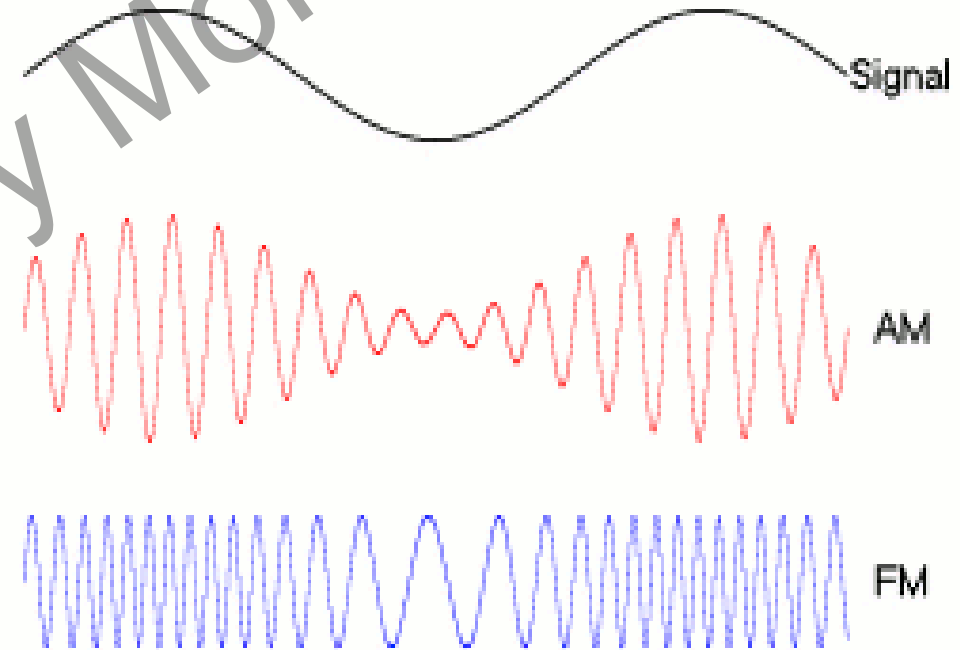


Baseband Vs Carrier Communication

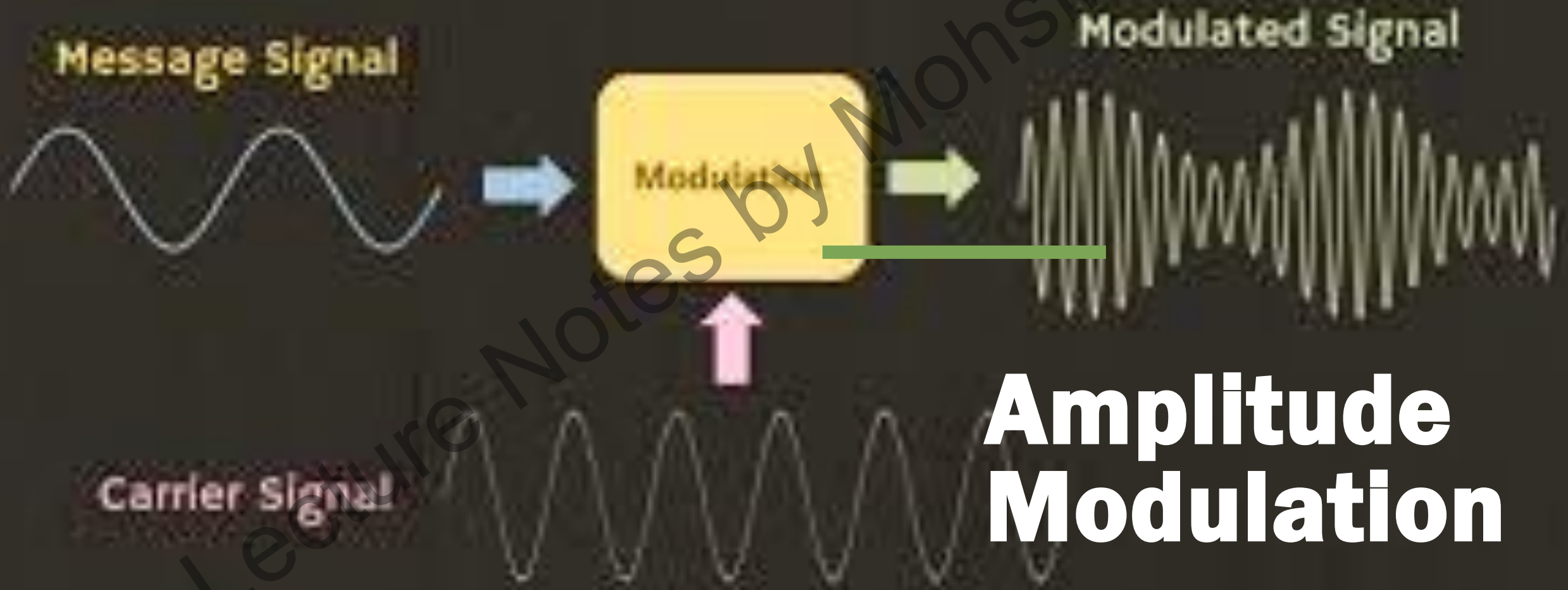


Carrier Communication

- One of the basic parameters (amplitude, frequency, or phase) of a **sinusoidal carrier** of **high frequency** ω_c is varied in proportion to the baseband signal $m(t)$.
- This results in amplitude modulation (AM), frequency modulation (FM), or phase modulation (PM).



Amplitude Modulation (AM)

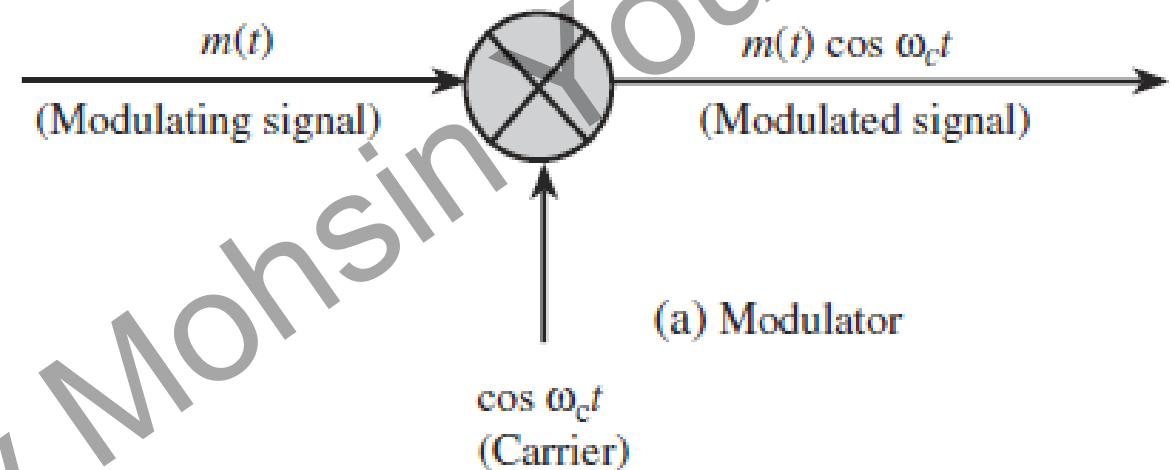


Amplitude Modulation

Double Sideband (DSB)

Amplitude 'A' of the **carrier**, $A \cos(\omega_c t + \theta_c)$ is varied in proportion to the baseband (**message**) signal $m(t)$, the **modulating signal**.

- The frequency ω_c and the phase θ_c are constant.
- If the carrier amplitude A is made directly proportional to the **modulating signal** $m(t)$, the **modulated signal** is $m(t) \cos \omega_c t$.



Thus, if

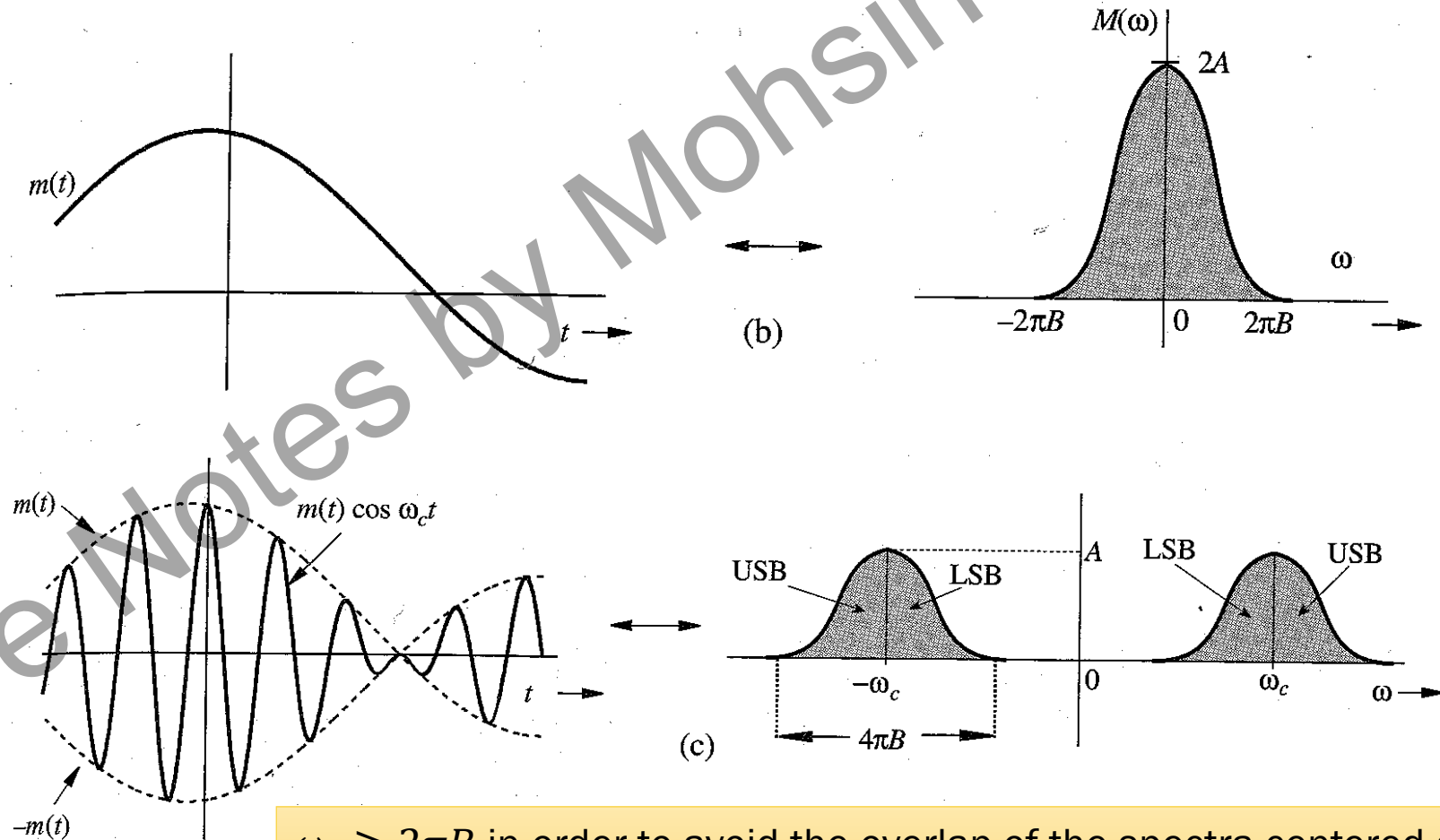
$$m(t) \Leftrightarrow M(\omega)$$

then

$$m(t) \cos \omega_c t \Leftrightarrow \frac{1}{2} [M(\omega + \omega_c) + M(\omega - \omega_c)]$$

Amplitude Modulation

Double Sideband Suppressed Carrier (DSB-SC)



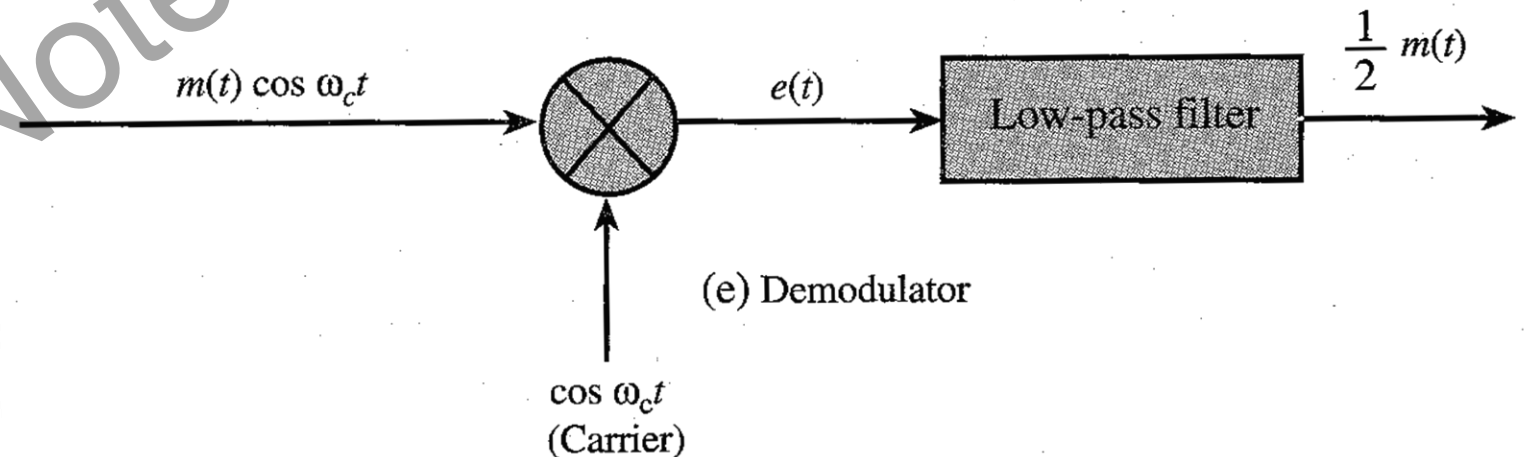
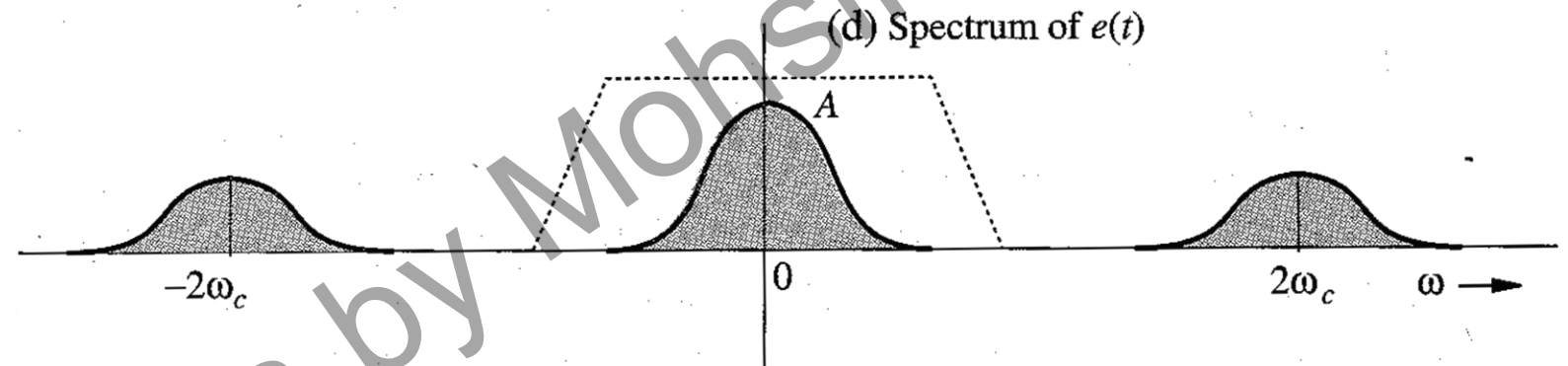
$\omega_c \geq 2\pi B$ in order to avoid the overlap of the spectra centered at $\pm\omega_c$

Demodulation of DSB-SC

The process of recovering the message signal $m(t)$ from the modulated signal $m(t) \cos \omega_c t$ (retranslating the spectrum to its original position) is referred to as **demodulation**, or **detection**.

$$\begin{aligned} e(t) &= m(t) \cos^2 \omega_c t \\ &= \frac{1}{2} [m(t) + m(t) \cos 2\omega_c t] \end{aligned}$$

Using the same carrier frequency (and phase) as used for modulation is called **synchronous (coherent) detection**.

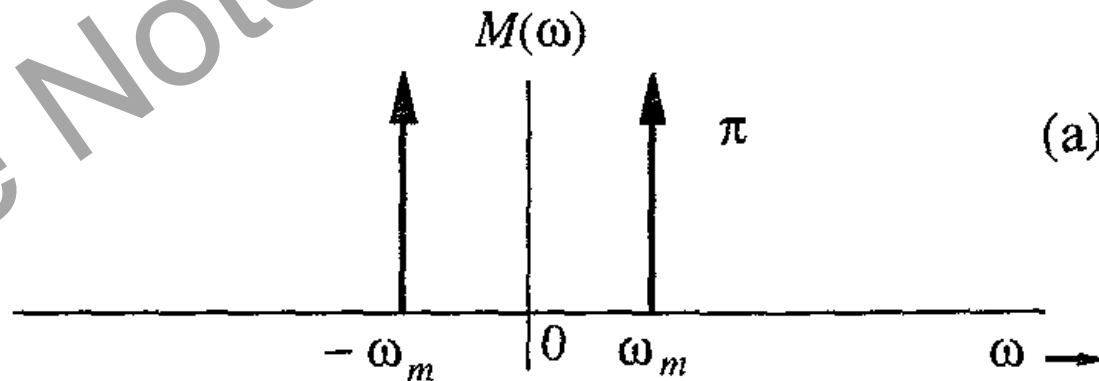
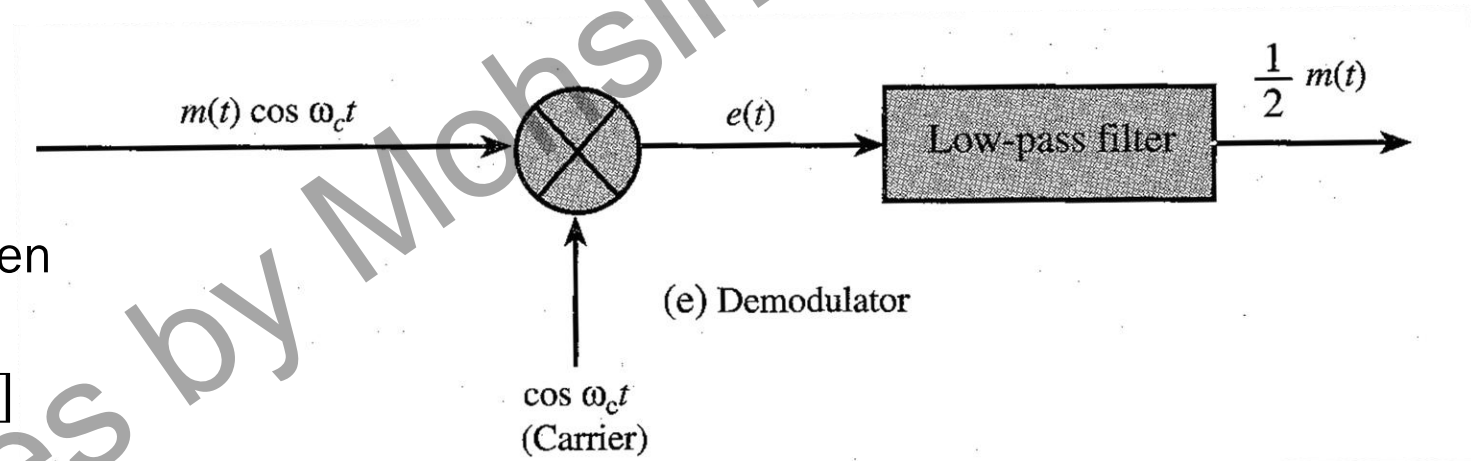


Example 4.1 Tone Modulation

For a baseband signal, $m(t) = \cos \omega_m t$, find the DSB-SC signal, and sketch its spectrum. Identify the USB and LSB. Verify that the DSB-SC modulated signal can be demodulated by the demodulator in Fig. 4.1e.

In the **frequency domain**, we work with signal spectra. The spectrum of the baseband signal $m(t) = \cos \omega_m t$ is given by:

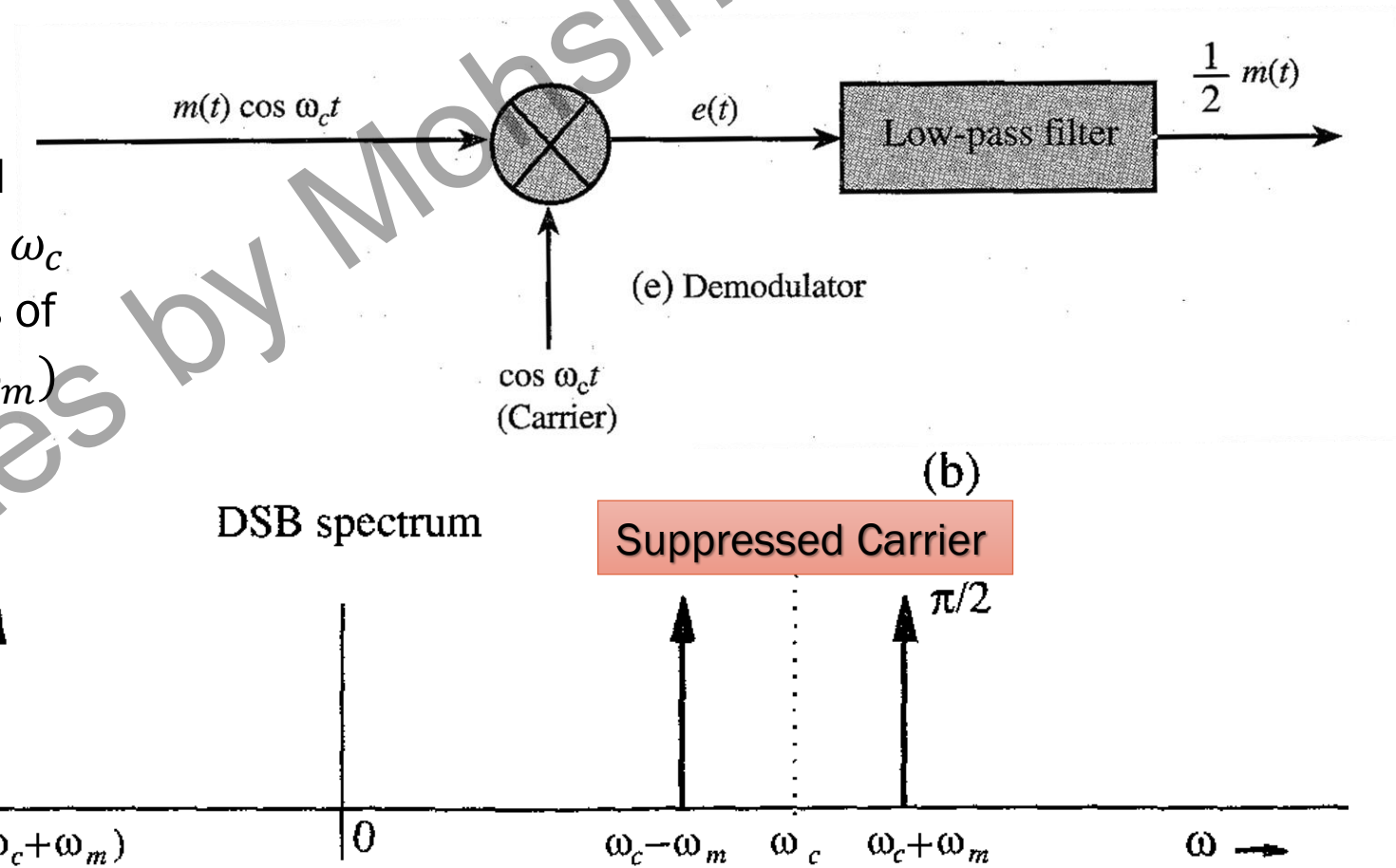
$$M(\omega) = \pi[\delta(\omega - \omega_m) + \delta(\omega + \omega_m)]$$



Example 4.1 Tone Modulation

For a baseband signal, $m(t) = \cos \omega_m t$, find the DSB-SC signal, and sketch its spectrum. Identify the USB and LSB. Verify that the DSB-SC modulated signal can be demodulated by the demodulator in Fig. 4.1e.

The DSB-SC spectrum of the baseband signal $m(t) = \cos \omega_m t$ is the baseband spectrum shifted to the right and left by ω_c (times one-half). This spectrum consists of impulses at $\pm(\omega_c - \omega_m)$ and $\pm(\omega_c + \omega_m)$

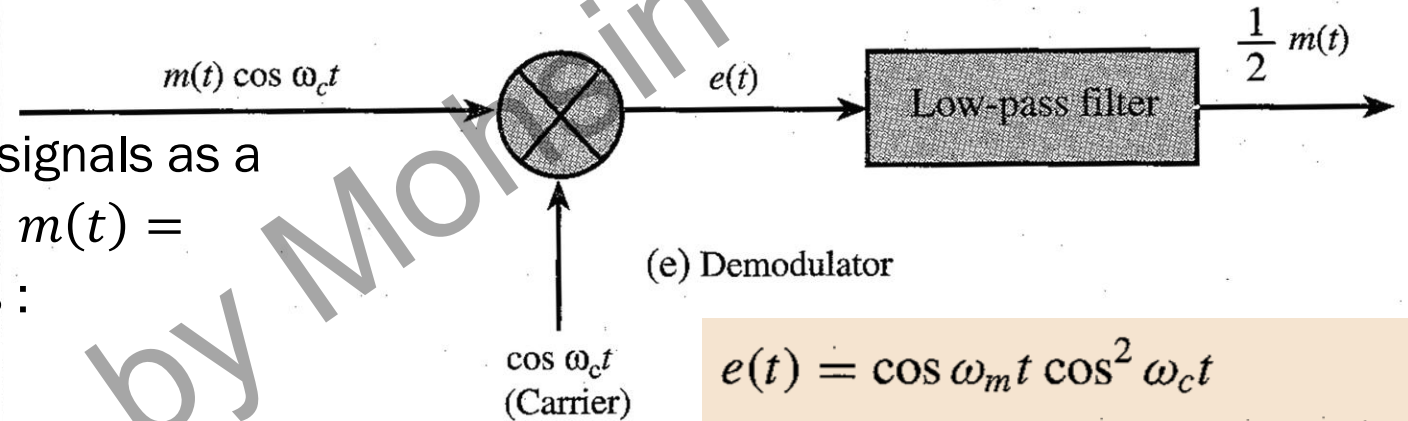


Example 4.1 Tone Modulation

For a baseband signal, $m(t) = \cos \omega_m t$, find the DSB-SC signal, and sketch its spectrum. Identify the USB and LSB. Verify that the DSB-SC modulated signal can be demodulated by the demodulator in Fig. 4.1e.

In the **time-domain**, we work directly with signals as a function of time. For the baseband signal $m(t) = \cos \omega_m t$, the DSB-SC signal $\varphi_{DSB-SC}(t)$ is :

$$\begin{aligned}\varphi_{DSB-SC}(t) &= m(t) \cos \omega_c t \\ &= \cos \omega_m t \cos \omega_c t \\ &= \frac{1}{2} [\cos(\omega_c + \omega_m)t + \cos(\omega_c - \omega_m)t]\end{aligned}$$



$$\begin{aligned}e(t) &= \cos \omega_m t \cos^2 \omega_c t \\ &= \frac{1}{2} \cos \omega_m t (1 + \cos 2\omega_c t)\end{aligned}$$

